Algorithm Sample 3

Dataset-expansion sample

1 Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

Initialize actor network μ_θ and critic networks Q_{ω_1} and Q_{ω_2}.
Initialize corresponding target networks: $\theta' \leftarrow \theta$, ${\omega'}_1 \leftarrow \omega_1$, ${\omega'}_2 \leftarrow \omega_2$ and choose $\rho_{\mathrm{p}} \in (0, 1)$.
Initialize replay buffer $\mathcal{R}$.
Choose a batch size K, a gradient based optimization algorithm and a corresponding learning rate $\lambda_{\text{actor}}, \lambda_{\text{critic}} > 0$ for both optimization procedures, a time step size Δt and a stopping criterion.
Choose standard deviation exploration noise σ_{expl} and lower and upper action bounds $a_{\text{low}}, a_{\text{high}}$.
repeat
    Select clipped action and step the environment dynamics forward.

$$a = \mathrm{clip}(\mu_\theta(s) + \epsilon, a_{\text{low}}, a_{\text{high}}), \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{expl}}).$$

    Observe next state s', reward r, and done signal d and store the tuple (s, a, r, s', d) in the replay buffer.
    if s' is terminal **then**
        Reset trajectory.
    end if
    for j in range(*update frequency*) **do**
        Sample batch $\mathcal{B} = \{(s^{(k)}, a^{(k)}, r^{(k)}, {s'}^{(k)}, d^{(k)})\}_{k=1}^{K}$ from replay buffer.
        Compute targets (Clipped Double Q-learning and policy smoothing).

$$y(r, s', d) = r + (1 - d) \min_{i=1,2} \left\{ Q_{{\omega'}_i}(s', \tilde{a}) \right\}, \quad \tilde{a} = \mathrm{clip}(\mu_{\theta'}(s) + \epsilon, a_{\text{low}}, a_{\text{high}}), \quad \epsilon \sim \mathcal{N}(0, \sigma_{\text{target}}).$$

        Estimate critic gradient $\nabla_\omega L(Q_\omega^{\mu_\theta})$ by

$$\nabla_{\omega_i} \left(\frac{1}{K} \sum_{k=1}^{K} \left(Q_{\omega_i}(s^{(k)}, a^{(k)}) - y(r^{(k)}, {s'}^{(k)}, d^{(k)}) \right)^2 \right), \quad \text{for } i = 1, 2.$$

        Update the critic parameters ω_i based on the optimization algorithm.
        if j mod *policy delay frequency* $= 0$ **then**
            Estimate actor gradient $\nabla_\theta J(\mu_\theta)$ by

$$\nabla_\theta \left(\frac{1}{K} \sum_{k=1}^{K} Q_{\omega_1}(s^{(k)}, \mu_\theta(s^{(k)})) \right).$$

            Update the actor parameters θ based on the optimization algorithm.

Update target networks softly:

$$\theta' \leftarrow \rho_{\mathrm{p}}\theta' + (1-\rho_{\mathrm{p}})\theta, \quad {\omega'}_i \leftarrow \rho_{\mathrm{p}}{\omega'}_i + (1-\rho_{\mathrm{p}})\omega_i, \quad \text{for } i = 1, 2.$$

end if

end for

until stopping criterion is fulfilled.